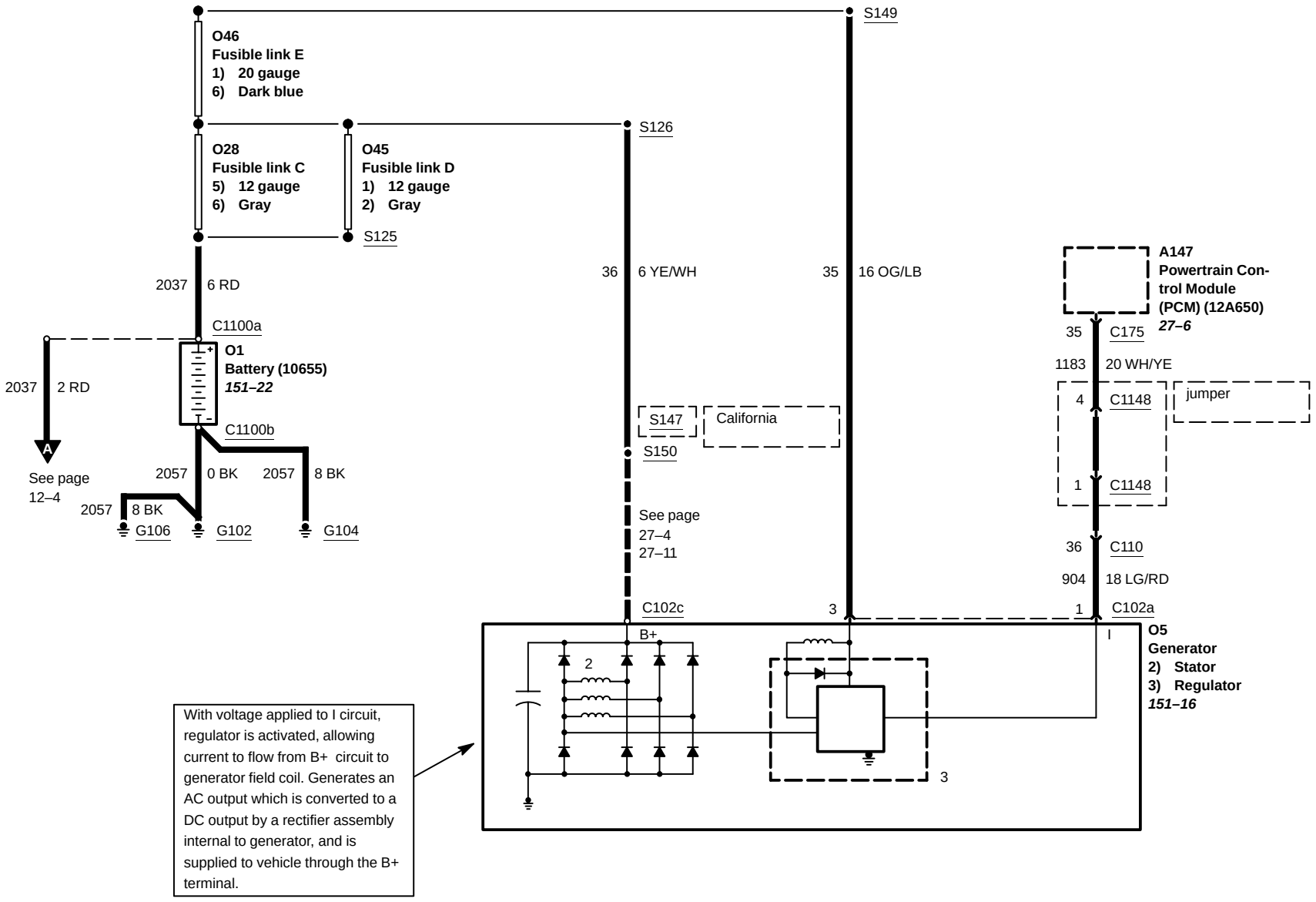Dual generators